CHANGE DETAILS

FROM	TO
Rev C	D: Revise statement in section 8 (remove option to stop video transmission)

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Scope

The scope of this document is to provide information about the fail-safe functionality of the product MAX96714. In the following chapters certain functionalities of the MAX96714 have been selected to be shown as an example. Further details can be reviewed as part of the Safety Case. The development of this product has been performed according to the ISO26262 standard.

1. Disclaimer

Maxim Integrated provides documentation for functional safety products. Within this documentation, Maxim Integrated has defined assumptions of use and operational boundary conditions. Based on these assumptions and boundary conditions, the capability of a product specific ASIL level has been assessed. Maxim's goal is to support customers to design and create their own end-product solutions that meet applicable functional safety standards and requirements. For this reason, Maxim Integrated does not guarantee ASIL compliance on a system level. The customer is fully responsible to perform the integration testing to confirm ASIL compliance on a system level. Maxim Integrated will not be liable for any claim or damage arising from customer's use of a Maxim Integrated product in a life support, life critical, or safety critical system, equipment, or application. Customer will indemnify, defend and hold Maxim Integrated harmless from and against any and all claims, damages, losses, costs, expenses, and liabilities arising out of the use of any Maxim Integrated product in any of these any such system, equipment or application. Maxim Integrated does not make any warranty as to the accuracy or completeness of this documentation and shall have no liability for its content.

2. Product description

The product MAX96714 is a dual GMSL2 compatible high-speed deserializer (GMSL=Gigabit Multimedia Serial Link) with dual CSI-2 outputs for automotive camera applications. It is used to transport video data from one or multiple remote cameras to an electronic control unit (ECU). The ECU is connected to the MAX96714 through a CSI-2 interface. One or more MAX96714 are typically mounted on the same board as the ECU.

The MAX96714 is used in combination with the MAX96717 or other GMSL2 serializers in order to transfer data via single ended coax cables or differential shielded twisted pairs. The high-speed serial input ports on MAX96714 can receive data in GMSL2 format on the two high speed serial inputs. The serializer/deserializer combination also supports a lower data rate reverse channel, which may be used to control the image sensors from the ECU, and to synchronize multiple cameras.

The supported intended applications are camera systems used for ADAS applications. Possible applications include forward looking and rear-view cameras for ADAS, virtual side mirrors and surround view camera systems.

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2.1. Product Features

The MAX96714 provides the following features.

- Up to 6Gb/s forward data rate and (187.5Mb/s) control channel
- 1 x 4-lane MIPI CSI-2 v1.3 ports
- D-PHY v1.2 at up to 2.5Gbps/lane
- 16 channel CSI-2 virtual channel support
- Support for all CSI-2 data types
- Supports CSI-2 forwarding or pixel mode video transport
- RoR – Reference over Reverse
- Programming through I2C or UART
- Line Fault Monitor
- Adaptive cable equalization
- Eye opening monitor (EOM) for real-time jitter margin measurement
- CSI-2 lane polarity flip and reassignment
- Concurrent control channel transactions with settable priority level
- Multi-function pins (MFP)
- 4 hardware-selectable device addresses
- 16-bit CRC protection of control channel data and 32-bit CRC protection of video data
- Retransmission of control channel data upon CRC error detection
- Spread spectrum

3. Item Definition

The product MAX96714/MAX96714F (GM16) is defined as an element/component and consists of parts which are identical to the individual functional blocks of the design. One or more elements grouped together build an Item. On the Item level the Hazard Analysis is performed, and the Functional Safety Concept is defined.

3.1. Item Architecture

The following diagram shows the item architecture describing the elements/components of the items and their parts. Furthermore, the interfaces between the elements and the interfaces to other items are described.

The item definition includes a description of the boundary with respect to the responsibility related to MAX96714/MAX96714F (as indicated with a dashed line in Fig. 1.). The consideration of Functional Safety outside this boundary is assumed to be in responsibility of the integrator.

The block diagram of the Item of product MAX96714/MAX96714F is shown in figure below. This includes the main control and data channels.

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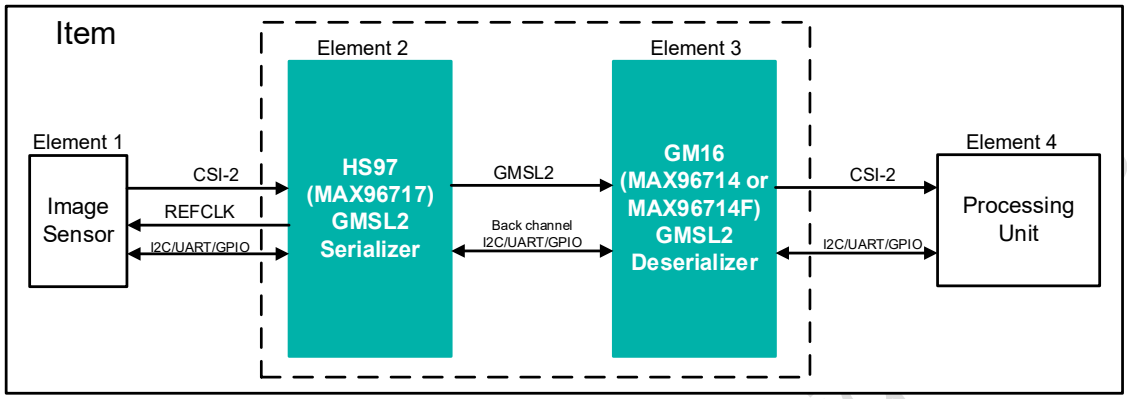


Figure 1: Item with the element MAX96714

3.1. System Definition

Video data is transmitted from an image sensor (Element 1) to the serializer (Element 2), where it is converted to GMSL2 serial link format. It is then transported across a cable and received on the high-speed serial input of the deserializer MAX96714/MAX96714F (Element 3). The deserializer MAX96714/MAX96714F converts the video to CSI-2 format. The CSI-2 data is forwarded to a processing unit (Element 4), which may be a CPU (Central processing Unit), a GPU (Graphics Processing Unit), an ISP (Image Signal Processor) or an FPGA (Field Programmable Gate Array).

The programming of the serializer MAX96717, deserializer MAX96714/MAX96714F and the image sensor, as well as status reporting from the image sensor and serializer is supported through the UART/I2C forward/backward control channels.

4. System Functions (Item Level)

The following system functions related to MAX96714 (Element 4) are defined at Item level.

System Function 1 **(SF1):** “Video transmission in Pixel Mode”

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Description: Receive and convert one or two streams high speed serial pixel data to parallel format. Optionally detect watermark. Add header and footer information to create CSI-2 packets. Aggregate and replicate data. Transmit the data on up to four CSI-2 ports.

System Function 2 **(SF2):** "Video transmission by CSI-2 Forwarding"
Description: Receive on GMSL inputs and convert one or two streams of high speed serially encoded CSI-2 packet data to parallel format. Aggregate and replicate CSI-2 packets. Transmit the CSI-2 packets on up to four CSI-2 ports.

System Function 3 **(SF3):** "Route control data over forward channel"
Description: Control data is received on the high-speed serial inputs. It is separated from the video data and is output on I2C, UART or GPIO pins.

System Function 4 **(SF4):** "Route control data over reverse channel"
Description: Transmit control data over the reverse channel. Control data is received on I2C, UART ports or GPIO for camera control and combined into the high speed (GMSL2) serial data streams.

5. Hazard Analysis (Item Level)

The Hazard Analysis is performed for each safety related System Function of Item.

5.1. Hazard Analysis for System Function SF1

The following Mal Functions (MF) are considered in combined situations with specific road conditions for System Function **SF1** "Video Transmission in Pixel Mode".

MF1: Error in reception of video data over high-speed serial link
MF2: Error in video data transmission over the CSI-2 Interface
MF3: Error in detection or insertion of watermark

The following road conditions are applicable for the Item under consideration and System Function SF1:
Road Condition RC1: Assisted Parking
Road Condition RC2: Stop and Go in the City
Road Condition RC3: Driving on Highway

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		Road Condition 1 RC1	Road Condition 2 RC2	Road Condition 3 RC3	Highest ASIL
		OM1.1 **Assisted Parking** (Object and Pedestrian Recognition activated, potential decision on steering and braking events)	**OM1.1** **Stop and Go in the City** (Object and Pedestrian Recognition activated, potential decision on emergency braking events, passengers crossing)	**OM1.1** **Driving on Highway** (Lane departure and forward collision warning and automatic braking activated, potential decision on braking or steering event)	**Related to Mal Function and Operating State**
Malfunction 1	Severity	**S2**	**S3**	**S3**	
	Exposure	**E3**	**E4**	**E4**	
Error in reception of video data over high-speed serial link	Controllability	**C2**	**C3**	**C3**	
Effect at Application Level: Wrong/corrupted/Missing Video Data, Consequence: wrong, missing or delayed object/passenger recognition No braking or steering event, when needed	ASIL-Combine	S2E3C2	S3E4C3	S3E4C3	
	ASIL Level	ASIL A	ASIL D	ASIL D	ASIL D
Malfunction 2	**Severity**	**S2**	**S3**	**S3**	
	Exposure	**E3**	**E4**	**E4**	
Error in video data transmission over the CSI-2 Interface	Controllability	**C2**	**C3**	**C3**	
Effect at Application Level: Wrong/corrupted/Missing Video Data, Consequence: wrong, missing or delayed object/passenger recognition No braking or steering event, when needed	ASIL-Combine	S2E3C2	S3E4C3	S3E4C3	
	ASIL	ASIL A	ASIL D	ASIL D	ASIL D

Table 1: Hazard Analysis for System Function SF1 (Video Transmission in Pixel Mode)

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5.1.1. Definition of Safety Goals

The definition of Safety Goals is based on the results of the Hazard Analysis. Safety Goals are defined for any malfunction which leads to an ASIL level and are summarized in the following table for SF1:

Safety Goal	Description	Malfunction	ASIL Level
SG1.1	Avoid reception and transmission of incorrect or delayed video data not corresponding to the source or delayed transmission	MF1, MF2, MF3	D
SG1.2	Avoid interruption in reception and transmission of video data	MF1, MF2, MF3	D

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5.2. Hazard Analysis for System Function SF2

The following Mal Functions (MF) are considered in combined situations with specific road conditions System Function SF2 “Video transmission by CSI-2 forwarding”.

MF1: Error in reception of video data over high-speed serial link
MF2: Error in video data transmission over the CSI-2 Interface

The following road conditions are applicable for the Item under consideration and System Function SF2:

Road Condition RC1: Assisted Parking
Road Condition RC2: Stop and Go in the City
Road Condition RC3: Driving on Highway

		Road - Condition 1 RC1	**Road - Condition 2 RC2**	**Road Condition 3 RC3**	**Highest ASIL**
		OM1.1 Assisted Parking (Object and Pedestrian Recognition activated, potential decision on steering and braking events)	**OM1.1 Stop and Go in the City** (Object and Pedestrian Recognition activated, potential decision on emergency braking events, passengers crossing)	**OM1.1 Driving on Highway** (Lane departure and forward collision warning and automatic braking activated, potential decision on braking or steering event)	**Related to Mal Function and Operating State**
Malfunction 1	Severity	**S2**	**S3**	**S3**	
	Exposure	**E3**	**E4**	**E4**	
Error in reception of video data over high-speed serial link	Controllability	**C2**	**C3**	**C3**	
Effect at Application Level: Wrong/corrupted/Missing Video Data, Consequence: wrong, missing or delayed object/passenger recognition No braking or steering event, when needed	ASIL-Combine	S2E3C2	S3E4C3	S3E4C3	
	ASIL Level	ASIL A	ASIL D	ASIL D	ASIL D
Malfunction 2	**Severity**	**S2**	**S3**	**S3**	
	Exposure	**E3**	**E4**	**E4**	
Error in video data transmission over the CSI-2 Interface	Controllability	**C2**	**C3**	**C3**	
Effect at Application Level: Wrong/corrupted/Missing Video Data, Consequence: wrong, missing or delayed object/passenger recognition No braking or steering event, when needed	ASIL-Combine	S2E3C2	S3E4C3	S3E4C3	

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	ASIL	ASIL A	ASIL D	ASIL D	ASIL D

Table 2: Hazard Analysis for System Function SF2 (Video transmission by CSI-2 Forwarding).

5.2.1. Definition of Safety Goals

The definition of Safety Goals is based on the results of the Hazard Analysis. Safety Goals are defined for any malfunction which leads to an ASIL level and are summarized in the following table for SF2:

Safety Goal	**Description**	**Malfunction**	**ASIL Level**
SG2.1	Avoid reception and transmission of incorrect or delayed video data not corresponding to the source or delayed transmission	MF1, MF2	D
SG2.2	Avoid interruption in reception and transmission of video data	MF1, MF2	D

5.3. Hazard Analysis for System Function SF3

The following Mal Functions (MF) are considered in combined situations with specific road conditions System Function SF3 “Route control data over forward channel”.

MF1: Error in reception of control data over high speed serial link
MF2: Error in output of control channel via UART/I2C, SPI and GPIO

The following road conditions are applicable for the Item under consideration and System Function SF3:
Road Condition RC1: Assisted Parking
Road Condition RC2: Stop and Go in the City
Road Condition RC3: Driving on Highway

Road - Condition 1 RC1	**Road - Condition 2 RC2**	**Road Condition 3 RC3**	**Highest ASIL**
OM1.1 Assisted Parking (Object and Pedestrian Recognition activated, potential decision on steering and braking events)	**OM1.1 Stop and Go in the City** (Object and Pedestrian Recognition activated, potential decision on emergency braking events, passengers crossing)	**OM1.1 Driving on Highway** (Lane departure and forward collision warning and automatic braking activated, potential decision on braking or steering event)	**Related to Mal Function and Operating State**

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Malfunction 1	**Severity**	**S2**	**S3**	**S3**	
	Exposure	**E3**	**E4**	**E4**	
Error in reception of control data over high speed serial link	Controllability	**C2**	**C3**	**C3**	
Effect at Application Level: Wrong/corrupted/Missing Control Data Consequence: Camera status may be incorrectly reported. Camera authentication may fail. Camera system may not start up.	ASIL-Combine	S2E3C2	S3E4C3	S3E4C3	
	ASIL	ASIL A	ASIL D	ASIL D	ASIL D
Malfunction 2	**Severity**	**S2**	**S3**	**S3**	
	Exposure	**E3**	**E4**	**E4**	
Error in output of control channel via UART/I2C and GPIO	Controllability	**C2**	**C3**	**C3**	
Effect at Application Level: Wrong/corrupted/Missing Control Data Consequence: Camera status may be incorrectly reported. Camera authentication may fail. Camera system may not start up.	ASIL-Combine	S2E3C2	S3E4C3	S3E4C3	
	ASIL	ASIL A	ASIL D	ASIL D	ASIL D

Table 3: Hazard Analysis for System Function SF3 (“Route control data over forward channel”).

5.3.1. Definition of Safety Goals

The definition of Safety Goals is based on the results of the Hazard Analysis. Safety Goals are defined for any malfunction which leads to an ASIL level and are summarized in the following table for SF3:

Safety Goal	Description	Malfunction	ASIL Level
SG3.1	Ensure that control data is received correctly from the GMSL link	MF1, MF2	D
SG3.2	Ensure that control data delivered on control channel pins is consistent with control channel data received on GMSL pins.	MF1, MF2	D

5.4. Hazard Analysis for System Function SF4

The following Mal Functions (MF) are considered in combined situations with specific road conditions System Function SF4 “Route control data over Reverse Channel”.

MF1: Error in reception of control data on UART/I2C, SPI and GPIO

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MF2: Error in output of control channel over high-speed serial link

The following road conditions are applicable for the Item under consideration and System Function SF4:

Road Condition RC1: Assisted Parking
Road Condition RC2: Stop and Go in the City
Road Condition RC3: Driving on Highway

		Road - Condition 1 RC1	**Road - Condition 2 RC2**	**Road Condition 3 RC3**	**Highest ASIL**
		OM1.1 Assisted Parking (Object and Pedestrian Recognition activated, potential decision on steering and braking events)	**OM1.1 Stop and Go in the City** (Object and Pedestrian Recognition activated, potential decision on emergency braking events, passengers crossing)	**OM1.1 Driving on Highway** (Lane departure and forward collision warning and automatic braking activated, potential decision on braking or steering event)	**Related to Mal Function and Operating State**
Malfunction 1	Severity	**S2**	**S3**	**S3**	
	Exposure	**E3**	**E4**	**E4**	
Error in reception of control data on UART/I2C and GPIO	Controllability	**C2**	**C3**	**C3**	
Effect at Application Level: Wrong/corrupted/Missing Control Data Consequence: Camera may be incorrectly configured, resulting in malformed or missing images. Camera authentication may not pass. Camera system may not start up.	ASIL-Combine	S2E3C2	S3E4C3	S3E4C3	
	ASIL Level	ASIL A	ASIL D	ASIL D	ASIL D
Malfunction 2	Severity	**S2**	**S3**	**S3**	
	Exposure	**E3**	**E4**	**E4**	
Error in output of control channel over high speed serial link	Controllability	**C2**	**C3**	**C3**	
Effect at Application Level: Wrong/corrupted/Missing Control Data Consequence: Camera may be incorrectly configured, resulting in malformed or missing images. Camera	ASIL-Combine	S2E3C2	S3E4C3	S3E4C3	

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authentication may not pass. Camera system may not start up.					
	ASIL Level	ASIL A	ASIL D	ASIL D	ASIL D

Table 3: Hazard Analysis for System Function SF4 ("Route control data over Reverse Channel").

Notes on Hazard Analysis: The following assumptions/justifications for the ratings in the Hazard Analysis are applicable.

1 Severity Ratings: A severity of S3 is assumed when the hazard effect is related to missing braking or steering events at medium to high speed. A severity of S2 is assumed when the hazard effect is related to missing braking or steering events at low speed.

2 Exposure ratings: Driving in the city or on the highway with cruise control and/or emergency brake functions activated is assumed to be E4. Using camera assisted systems for parking is assumed to be E3. 3

3 Controllability ratings: Sudden fail of cruise control systems leading to unintended or missing steering or braking events is assumed to be a situation which is difficult to control by the driver and therefore set to C3

5.4.1. Definition of Safety Goals

The definition of Safety Goals is based on the results of the Hazard Analysis. Safety Goals are defined for any malfunction which leads to an ASIL level and are summarized in the following table for SF4:

Safety Goal	Description	Malfunction	ASIL Level
SG4.1	Ensure that control data is received correctly on multi-function I/O prior to executing any commands or delivery of data to GMSL link	MF1, MF2	D
SG4.2	Ensure that control data is delivered correctly across the GMSL link reverse channel	MF2, MF2	D

6. ASIL Decomposition from item to element level

No ASIL decomposition is applied at the IC level.

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7. Functional Safety Concept (Item Level)

The Functional Safety Concept for Element MAX96714 with respect to the Item Definition and Hazard Analysis is specified for the following System Functions:

System Function 1 (**SF1**): "Video transmission in Pixel mode"
System Function 2 (**SF2**): "Video Transmission by CSI-2 Forwarding"
System Function 3 (**SF3**): "Route Control Data Over Forward Channel"
System Function 4 (**SF4**): "Route Control Data Over Reverse Channel"

The Functional Safety Concept for MAX96714 with the functional break down of System Functions to Element Functions and Component Functions of MAX96714 is shown in the following table:

Item Level Hazard Analysis					Element Level		
System Function	**ASIL**	**Safety Goal**	**Fault Tolerant Time [ms]**	**Safe State**	**Element and/or Component Function**	**ASIL**	**Fault Tolerant Time [ms]**
SF1	D	SG1.1 SG1.2	15	1. Error flagged on pin ERRB or LOCK 2. Dedicated register set (UART/I2C)	EF1: Receive high speed serial data on GMSL port	B(*1)	15
					EF2: Convert serial data to parallel data	B(*1)	15
					EF3: Decode received data and strip off line code	B(*1)	15
					EF4: Detect, correct and report data errors	B(*1)	15
					EF5: Descramble data	B(*1)	15
					EF6: De-packetize, split out video data. Route video data	B(*1)	15
					EF7: Detect or insert watermark	B(*1)	15
					EF8: Aggregate pixel data streams	B(*1)	15
					EF9: Convert pixel stream to CSI-2 packet data	B(*1)	15
					EF10: Replicate and route CSI-2 data	B(*1)	15
					EF11: Output CSI-2 data	B(*1)	15

Table 4: Functional Safety Concept, System Function SF1 for MAX96714

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Item Level Hazard Analysis					Element Level		
System Function	ASIL	Safety Goal	Fault Tolerant Time [ms]	Safe State	Element and/or Component Function	ASIL	Fault Tolerant Time [ms]
SF2	D	SG 2.1 SG2.2	15	1. Error flagged on pin ERRB or LOCK 2. Dedicated register set (UART/I2C) 3. End-to-end CRC fails	EF1: Receive high speed serial data on GMSL port	B(*1)	15
					EF2: Convert serial data to parallel data	B(*1)	15
					EF3: Decode received data and strip off line code	B(*1)	15
					EF4: Detect, correct and report data errors	B(*1)	15
					EF5: Descramble data	B(*1)	15
					EF6: De-packetize, split out video data. Route video data	B(*1)	15
					EF12: Bypass watermark block	B(*1)	15
					EF13: Aggregate and forward CSI-2 data	B(*1)	15
					EF10: Replicate and route CSI-2 data	B(*1)	15
					EF11: Output CSI-2 data	B(*1)	15

Table 5: Functional Safety Concept, System Function SF2 for MAX96714

Item Level Hazard Analysis					Element Level		
System Function	ASIL	Safety Goal	Fault Tolerant Time [ms]	Safe State	Element and/or Component Function	ASIL	Fault Tolerant Time [ms]
SF3	D	SG 3.1 SG3.2	15	1. Error detection and retransmission of command during fault tolerant time 2. End-to-end CRC on control channel data fails 3. Error indication on ERRB 4. Dedicated register is set	EF1: Receive high speed serial data on GMSL port	B(*1)	15
					EF2: Convert serial data to parallel data	B(*1)	15
					EF3: Decode received data and strip off line code	B(*1)	15
					EF4: Detect, correct and report data errors	B(*1)	15
					EF5: Descramble data	B(*1)	15
					EF21: De-packetize and route control data	B(*1)	15
					EF14: Decode control data in adapters	B(*1)	15
					EF15: Output data on multi-function I/O	B(*1)	15

Table 6: Functional Safety Concept, System Function SF3 for MAX96714

Item Level Hazard Analysis					Element Level		
System Function	ASIL	Safety Goal	Fault Tolerant Time [ms]	Safe State	Element and/or Component Function	ASIL	Fault Tolerant Time [ms]

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SF4	D	SG 4.1 SG4.2	15	1. Error detection and retransmission of command during fault tolerant time 2. End-to-end CRC on control channel data fails 3. Error indication on ERRB 4. Dedicated register is set	EF16: Receive control data on multi-function pins	B(*1)	15
					EF17: Buffer control data, generate and schedule GMSL2/3 packets	B(*1)	15
					EF18: Scramble and encode data	B(*1)	15
					EF19: Serialize data	B(*1)	15
					EF20: Transmit serial data onto link	B(*1)	15

Table 7: Functional Safety Concept, System Function SF4 for MAX96714

Note (*1): The IC supports ASIL B for the related Safety Goals. If higher ASIL ratings should be achieved at system level, ASIL decomposition needs to be performed: ASIL D = ASIL B (D) + ASIL B (D)

SG1.1: Avoid transmission of incorrect video data not corresponding to the source or delayed transmission in pixel mode
SG1.2: Avoid interruption in reception and transmission of video data in pixel mode
SG2.1: Avoid reception and transmission of incorrect video data not corresponding to the source or delayed transmission in CSI-2 forwarding mode
SG2.2: Avoid interruption in reception and transmission of video data in CSI-2 forwarding mode
SG3.1: Ensure that control data is received correctly from the GMSL link
SG3.2: Ensure that control data delivered to control channel pins is consistent with control channel data received on GMSL pins.
SG4.1: Ensure that control data is received correctly on multi-function I/O prior to executing any commands or delivery of data to GMSL link
SG4.2: Ensure that control data is delivered correctly across the GMSL link reverse channel

8. Technical Safety Concept

Technical Safety Concept specifies how the required functionality of the IC is implemented. Special focus is on diagnostics and safety measures.

In order to realize the functional safety concept using a dedicated hardware implementation, the technical safety concept is described as:

Video and control data transmission errors are detected through built-in safety mechanisms. Upon error detection, the fault status is reported to external elements. Status registers provide detailed fault reporting.

The specific Technical Safety Requirements with respect to Safety Mechanisms for detection of internal failures (ISO26262-4/6.4.2.2) for MAX96714 are specified in the table below for all element functions (EF1, EF2, …). Those definitions contain as relevant information the description of Safety Measures on all potential failures in the used circuit blocks contributing to realize a specific element function, the diagnostic limit window (range for issuing an alarm), the diagnostic test time and the reaction on a fault condition. The allocation of functional blocks to the element functions is described. The following table summarizes the Technical Safety Requirements for MAX96714.

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Element Function/ System Function	Functional Blocks	Description Functional Blocks	Faults to be Detected*	Safety Measures	Description	DTI (Diagnostic Test Time)	Fault Reaction State
EF1/SF1	FB 1	Serial data receive amplifier	Incorrect gain	SM1, SM3, SM4, SM6, SM14	BIST for serial link CRC for control channel CRC for forward video channel Lock indicator for forward channel Eye Opening Monitor	10 ms	ERRB pin asserted, dedicated register set LOCK pin deasserted
			Incorrect input impedance	SM1, SM3, SM4, SM6, SM14	BIST for serial link CRC for control channel CRC for forward video channel Lock indicator for forward channel Eye Opening Monitor	10 ms	ERRB pin asserted, dedicated register set LOCK pin deasserted
	FB 2	Adaptive equalizer and timing recovery	Incorrect equalization	SM1, SM3, SM4, SM6, SM14	BIST for serial link CRC for control channel CRC for forward video channel Lock indicator for forward channel Eye Opening Monitor	10 ms	ERRB pin asserted, dedicated register set LOCK pin deasserted
			Incorrect sampling point	SM1, SM3, SM4, SM6, SM14	BIST for serial link CRC for control channel CRC for forward video channel Lock indicator for forward channel Eye Opening Monitor	10 ms	ERRB pin asserted, dedicated register set LOCK pin deasserted
			Incorrect data slicing	SM1, SM3, SM4, SM6, SM14	BIST for serial link CRC for control channel CRC for forward video channel Lock indicator for forward channel Eye Opening Monitor	10 ms	ERRB pin asserted, dedicated register set LOCK pin deasserted
EF2/SF1	FB 3	Serial to Parallel	Incorrect conversion to parallel data	SM1, SM3, SM4, SM6, SM32	BIST for serial link CRC for control channel CRC for forward video channel Lock indicator for forward channel Running disparity/illegal 9b/10b symbol check	10 ms	ERRB pin asserted, dedicated register set LOCK pin deasserted
EF3/SF1	FB 4	9b/10b decoder	Incorrect decoding	SM1, SM3, SM4, SM6, SM32	BIST for serial link CRC for control channel CRC for forward video channel Lock indicator for forward channel Running disparity/illegal 9b/10b symbol check	10 ms	ERRB pin asserted, dedicated register set LOCK pin deasserted
EF5/SF1	FB 7	Descrambler	Incorrect descrambling	SM1, SM3, SM4	BIST for serial link CRC for control channel CRC for forward video channel	10 ms	ERRB pin asserted, dedicated register set
EF6/SF1	FB 8	Split video and forward control data. De-packetize data	Incorrect data routing	SM4, SM3, SM1	CRC for forward video channel, CRC for control channel Serial link BIST	10 ms	ERRB pin asserted, dedicated register set
			Incorrect de-packetization of video data	SM4, SM1	CRC for forward video channel, Serial link BIST	10 ms	ERRB pin asserted, dedicated register set
EF8/SF1	FB 10	Line buffer	Stuck memory bit (permanent and transient)	SM26, SM19	Memory ECC Memory BIST on startup	10 ms	ERRB pin asserted, dedicated register set
	FB10	Line Buffer	Read/write pointer incorrect	SM19 SM17 SM36 SM18	Memory BIST on startup FIFO overflow detection Video lock Logic BIST	10 ms	ERRB pin asserted, dedicated register set

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	FB11	Tunnel/Pixel Mode Data Routing	CSI-2 data routed to incorrect controller	SM36	Video lock	10 ms	ERRB pin asserted, dedicated register set
EF9/SF1	FB 13	CSI-2 Tx controller	Incorrect conversion from pixel data to CSI-2 packet	SM35 SM18	CSI-2 to D-PHY handshake Logic BIST	10 ms	ERRB pin asserted, dedicated register set
EF11/SF1	FB 19	CSI-2 PHY	Stuck output	SM31	CSI-2 CRC checked at host. Interrupt provided to deserializer	10 ms	ERRB pin asserted, dedicated register set

Element Function/ System Function	Functional Blocks	Description Functional Blocks	Faults to be Detected*	Safety Measures	Description	DTI	Fault Reaction State
EF1/SF2	FB 1	Serial data receive amplifier	Incorrect gain	SM1, SM3, SM4, SM6, SM14 SM20, SM31	BIST for serial link CRC for control channel CRC for forward video channel Lock indicator for forward channel Eye Opening Monitor End-to-end CRC for CSI-2 (*1) Exterior triggering of interrupt (*1)	10 ms	ERRB pin asserted, dedicated register set LOCK pin deasserted
			Incorrect input impedance	SM1, SM3, SM4, SM6, SM14. SM20, SM31	BIST for serial link CRC for control channel CRC for forward video channel Lock indicator for forward channel Eye Opening Monitor End-to-end CRC for CSI-2(*1) Exterior triggering of interrupt (*1)	10 ms	ERRB pin asserted, dedicated register set LOCK pin deasserted
	FB 2	Adaptive equalizer and timing recovery	Incorrect equalization	SM1, SM3, SM4, SM6, SM14, SM20, SM31	BIST for serial link CRC for control channel CRC for forward video channel Lock indicator for forward channel Eye Opening Monitor End-to-end CRC for CSI-2 (*1) Exterior triggering of interrupt (*1)	10 ms	ERRB pin asserted, dedicated register set LOCK pin deasserted
			Incorrect sampling point	SM1, SM3, SM4, SM6, SM14, SM20, SM31	BIST for serial link CRC for control channel CRC for forward video channel Lock indicator for forward channel Eye Opening Monitor End-to-end CRC for CSI-2 (*1) Exterior triggering of interrupt (*1)	10 ms	ERRB pin asserted, dedicated register set LOCK pin deasserted
			Incorrect data slicing	SM1, SM3, SM4, SM6, SM14, SM20, SM31	BIST for serial link CRC for control channel CRC for forward video channel Lock indicator for forward channel Eye Opening Monitor End-to-end CRC for CSI-2 (*1) Exterior triggering of interrupt (*1)	10 ms	ERRB pin asserted, dedicated register set LOCK pin deasserted
EF2/SF2	FB 3	Serial to Parallel	Incorrect conversion to parallel data	SM1, SM3, SM4, SM6, SM32 SM20, SM31	BIST for serial link CRC for control channel CRC for forward video channel Lock indicator for forward channel Running disparity/illegal 9b/10b symbol check End-to-end CRC for CSI-2 (*1)	10 ms	ERRB pin asserted, dedicated register set

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					Exterior triggering of interrupt (*1)		
EF3/SF2	FB 4	9b/10b decoder	Incorrect decoding	SM1, SM3, SM4, SM6, SM32 SM20 , SM31	BIST for serial link CRC for control channel CRC for forward video channel Lock indicator for forward channel Running disparity/illegal 9b/10b symbol check End-to-end CRC for CSI-2 (*1) Exterior triggering of interrupt (*1)	10 ms	ERRB pin asserted, dedicated register set LOCK pin deasserted
EF5/SF2	FB 7	Descrambler	Incorrect descrambling	SM1, SM3, SM4 SM20, SM31	BIST for serial link CRC for control channel CRC for forward video channel End-to-end CRC for CSI-2 (*1) Exterior triggering of interrupt (*1)	10 ms	ERRB pin asserted, dedicated register set
EF6/SF2	FB 8	Split video and forward control data. De-packetize data	Incorrect data routing	SM4, SM3, SM1 SM20, SM31	CRC for forward video channel, CRC for control channel Serial link BIST End-to-end CRC for CSI-2 (*1) Exterior triggering of interrupt (*1)	10 ms	ERRB pin asserted, dedicated register set
			Incorrect de-packetization of video data	SM4, SM1 SM20, SM31	CRC for forward video channel, Serial link BIST End-to-end CRC for CSI-2 (*1) Exterior triggering of interrupt (*1)	10 ms	ERRB pin asserted, dedicated register set
EF13/SF2	FB 10	Line buffer	Stuck memory bit	SM4, SM6, SM1 SM20, SM31	CRC for forward video channel, LOCK indicator Serial link BIST End-to-end CRC for CSI-2 (*1) Exterior triggering of interrupt	10 ms	ERRB pin asserted, dedicated register set
	FB 10	Line buffer	Read/write pointer incorrect	SM20, SM31	End-to-end CRC for CSI-2 Exterior triggering of interrupt (*1)	10 ms	ERRB pin asserted, dedicated register set
	FB11	Tunnel/Pixel Mode Data Routing	CSI-2 data routed to incorrect controller	SM20, SM31	End-to-end CRC for CSI-2 (*1) Exterior triggering of interrupt (*1)	10 ms	ERRB pin asserted, dedicated register set
EF9/SF2	FB 12	CSI-Tunnel	Error in data forwarding	SM20, SM31	End-to-end CRC for CSI-2 (*1) Exterior triggering of interrupt (*1)	10 ms	ERRB pin asserted, dedicated register set
EF11/SF2	FB 19	CSI-2 PHY	Corrupt video data	SM20, SM31	End-to-end CRC for CSI-2 (*1) Exterior triggering of interrupt (*1)	10 ms	ERRB pin asserted, dedicated register set
			Corrupt CSI-2 header or footer	SM20, SM31	End-to-end CRC for CSI-2 (*1) Exterior triggering of interrupt (*1)	10 ms	ERRB pin asserted, dedicated register set

(*1) SM20 and SM31 are evaluated at application level

Element Function/ System Function	Functional Blocks	Description Functional Blocks	Faults to be Detected*	Safety Measures	Description	DTI	Fault Reaction State
EF1/SF3	FB 1	Serial data receive amplifier	Incorrect gain	SM1, SM3, SM5, SM6, SM7, SM14	BIST for serial link CRC for control channel, Retransmission on control channel Lock indicator of forward channel, ERRB indicator of forward channel Eye Opening Monitor	10 ms	ERRB pin asserted, dedicated register set LOCK pin deasserted

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			Incorrect input impedance	SM1, SM3, SM5, SM6, SM7, SM14	BIST for serial link CRC for control channel, Retransmission on control channel Lock indicator of forward channel, ERRB indicator of forward channel Eye Opening Monitor	10 ms	ERRB pin asserted, dedicated register LOCK pin deasserted r set
	FB 2	Adaptive equalizer and timing recovery	Incorrect equalization	SM1, SM3, SM5, SM6, SM7, SM14	BIST for serial link CRC for control channel, Retransmission on control channel Lock indicator of forward channel, ERRB indicator of forward channel Eye Opening Monitor	10 ms	ERRB pin asserted, dedicated register set LOCK pin
			Incorrect sampling point	SM1, SM3, SM5, SM6, SM7, SM14	BIST for serial link CRC for control channel, Retransmission on control channel Lock indicator of forward channel, ERRB indicator of forward channel Eye Opening Monitor	10 ms	ERRB pin asserted, dedicated register set
			Incorrect data slicing	SM1, SM3, SM5, SM6, SM7, SM14	BIST for serial link CRC for control channel, Retransmission on control channel Lock indicator of forward channel, ERRB indicator of forward channel, Eye Opening Monitor	10 ms	ERRB pin asserted, dedicated register set, LOCK pin deasserted
EF2/SF3	FB 3	Serial to Parallel	Incorrect conversion to parallel data	SM1, SM3, SM5, SM6, SM7	BIST for serial link CRC for control channel, Retransmission on control channel Lock indicator of forward channel, ERRB indicator of forward channel	10 ms	ERRB pin asserted, dedicated register set, LOCK pin deasserted
EF3/SF3	FB 4	9b/10b decoder	Incorrect decoding	SM1, SM3, SM5, SM6, SM7	BIST for serial link CRC for control channel, Retransmission on control channel Lock indicator of forward channel, ERRB indicator of forward channel	10 ms	ERRB pin asserted, dedicated register set LOCK pin deasserted
EF5/SF3	FB 7	Descrambler	Incorrect descrambling	SM1, SM3, SM5, SM7	BIST for serial link CRC for control channel, Retransmission on control channel ERRB indicator of forward channel	10 ms	ERRB pin asserted, dedicated register set
EF21/SF3	FB8	Rx Base Band: Split video and forward Control Data. De-packetize data	Stuck data bit	SM3, SM5, SM7, SM18,	CRC for control channel, Retransmission on control channel ERRB indicator of forward channel Logic BIST	10 ms	ERRB pin asserted, dedicated register set
			Incorrect routing of data/stuck control bit	SM18	Logic BIST	10 ms	ERRB pin asserted, dedicated register set
EF14/EF3	FB20	GPIO	Incorrect routing of GPIO data/stuck control bit	SM18	Logic BIST	10 ms	ERRB pin asserted, dedicated register set
			Incorrect GPIO driver output	SM23	GPO readback	10 ms	ERRB pin asserted, dedicated register set
	FB21	I2C/UART host interface	I2C/UART data stuck I2C message loss or insertion	SM12, SM21, SM22, SM18	ACK on I2C or UART CRC on I2C and UART transactions Message counter Logic BIST	10 ms	ERRB pin asserted, dedicated register set
	FB23	I2C/UART Tunnels	I2C/UART data stuck	SM18	Logic BIST Note: For I2C/UART tunnel, CRC is evaluated at application level	10 ms	ERRB pin asserted, dedicated register set
EF15	FB24	Map to multi-function I/O	Incorrect data routing	SM18, SM23	Logic BIST GPO Readback	10 ms	ERRB pin asserted,

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							dedicated register set
			Stuck output driver	SM23	GPO Readback	10 ms	ERRB pin asserted, dedicated register set
			Open output driver	SM23	GPO Readback	10 ms	ERRB pin asserted, dedicated register set

Element Function/ System Function	Functional Blocks	Description Functional Blocks	Faults to be Detected*	Safety Measures	Description	DTI	Fault Reaction State
EF16/SF4	FB 24	Multi-function I/O and mapper	Incorrect data routing/stuck control bit	SM18	Logic BIST	10 ms	ERRB pin asserted and dedicated register set
			Stuck data bit	SM18 SM23 SM37	Logic BIST GPO readback GPIO open detection	10 ms	ERRB pin asserted and dedicated register set
	FB21	I2C/UART host interface	Corrupted, missing or duplicated message	SM21, SM22	CRC check on I2C and UART transactions Message counter for I2C and UART	10 ms	Message NACKed
	FB23	I2C/UART tunnel	Corrupted message	SM18	Logic BIST Note: CRC evaluated at application level	10 ms	POST failed
	FB20	GPIO	Stuck data bit	SM18 SM37	Logic BIST GPIO open detection	10 ms	POST failed
	FB25	Line fault detector	Stuck output	SM18	Logic BIST	10 ms	POST failed
EF17/SF4	FB26	Scheduler	Incorrect generation of packets	SM18, SM3, SM5	Logic BIST CRC for control channel Automatic retransmission	10 ms	ERRB pin asserted, dedicated register set POST failed
EF18/SF4	FB27	Scrambler	Incorrect scrambling	SM1,SM3, SM5, SM6, SM18	BIST for serial link CRC for control channel Automatic retransmission LOCK indicator of forward channel Logic BIST	10 ms	ERRB pin asserted, dedicated register set POST failed LOCK pin deasserted
		9b/10b encoder	Incorrect encoding	SM1, SM3, SM5, SM6, SM18	BIST for serial link CRC for control channel Automatic retransmission LOCK indicator of forward channel Logic BIST	10 ms	ERRB pin asserted, dedicated register set POST failed LOCK pin deasserted
EF19/SF4	FB 28	Parallel to serial conversion	Incorrect serialization	SM1, SM3, SM5, SM6, SM18	BIST for serial link CRC for control channel Automatic retransmission LOCK indicator of forward channel Logic BIST	10 ms	ERRB pin asserted, dedicated register set POST failed LOCK pin deasserted
EF20/SF4	FB 1	Driver amplifier	Output stuck at 1 or 0	SM1, SM3, SM5, SM6, SM18, SM14	BIST for serial link CRC for control channel Automatic retransmission LOCK indicator of forward channel	10 ms	ERRB pin asserted, dedicated register set

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					Logic BIST		POST failed LOCK pin deasserted
			Too low output power	SM1, SM3, SM5, SM6, SM14	BIST for serial link CRC for control channel Automatic retransmission LOCK indicator of forward channel Eye opening monitor	10 ms	ERRB pin asserted, dedicated register set POST failed LOCK pin deasserted
	FB17	Device configuration and control	CFG pin decoding error causing incorrect CXTP or MAX96714F GMSL1 setting	SM1, SM3, SM5, SM6, SM14FB1	BIST for serial link CRC for control channel Automatic retransmission LOCK indicator of forward channel Eye opening monitor	10 ms	ERRB pin asserted, dedicated register set POST failed LOCK pin deasserted

Table 8: Technical Safety Requirements for MAX96714

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8.1. Hardware Design Specification for MAX96714

The following schematics shows the implementation of the Hardware Architecture including all the relevant blocks in the design according to the block diagram. The naming of the blocks is identical to the description in Table 1.

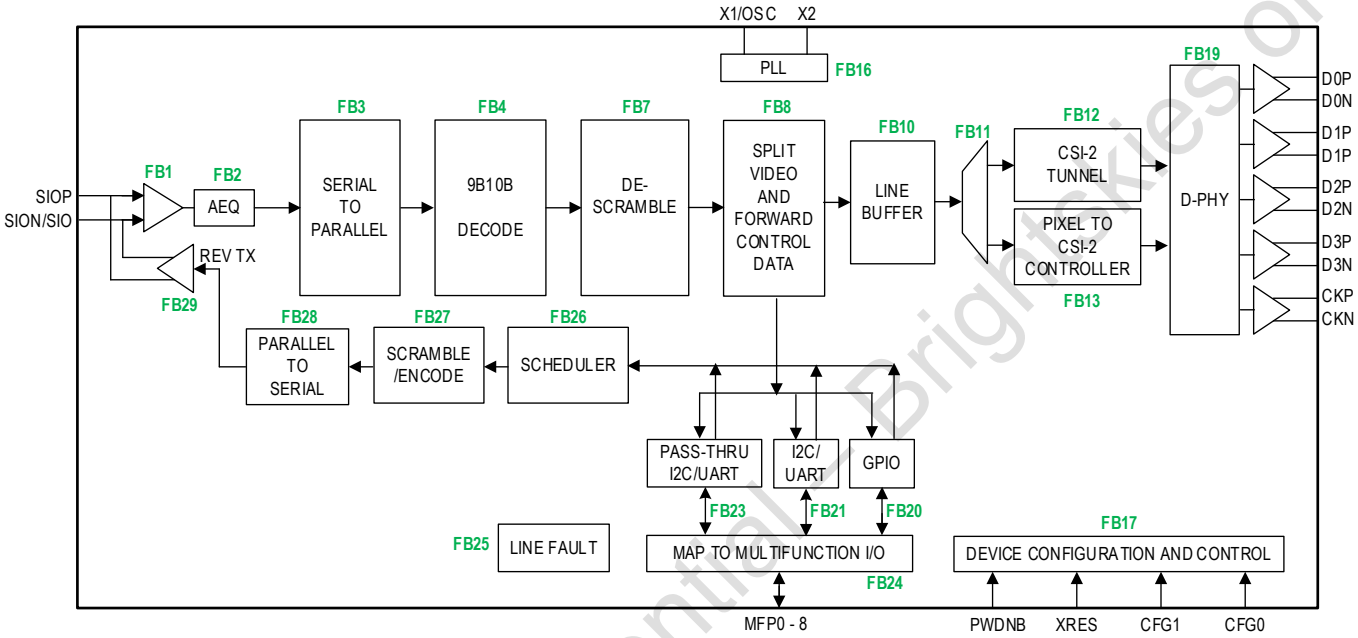


Figure 2: Hardware Design Specification/Block Diagram of MAX96714

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8.2. List of Safety Measures for MAX96714

The list of defined Safety Measures for MAX96714 as part of the Technical Safety Concept are described in the table below.

SM	Safety Mechanism (Name and Description)	Fault to be detected	Diagnostic Test Interval	Operational Mode	Optimized SM Default/Optional		Routed to ERRB
					Pixel Mode	Tunnel Mode	
SM1	**BIST for Serial Link (BER Testing using PRBS):** Built in PRBS Checker for BER Testing of the full-duplex Serial Link for forward and reverse direction	Data Corruption – Video Channel, Bit Error Rate, Link Issues	<10us	startup	Optional	Optional	Yes (decode/idle err) and No (link PRBS synced)
SM2	**Line fault detection on Serializer Output:** Line faults (short to Batt/Gnd or open) are detected on the GMSL 2 output pins TX0+/- and TX1+/-.x .The result is flagged on the LFLTB pin.	Line Faults on the Serial Link Output (Short to Bat/Gnd/Open)	<10us	continuous	Optional	Optional	Yes (latched LFLT not enabled by default)
SM3	**CRC for Control Channel** (UART/I2C/GPIO/SPI) in Forward and Reverse Mode, 16 BIT Packet CRC	Data corruption on the control channel	<10us (forward), <100us (reverse)	continuous	Default	Default	Yes
SM4	**CRC for Forward Video Channel** (32 BIT VIDEO LINE CRC AND 16 BIT PACKET CRC)	Data corruption on the forward video channel	<50us	continuous	Default	Default	Yes
SM5	**Error Correction through Re-transmission on Control Channel** (UART/I2C/GPIO/SPI) in Forward and Reverse Mode. Correction of up to 16 Bit Errors in each packet.	Re-transmission in case of: -incorrect CRC -Dropped packets	<10us (forward), <100us (reverse)	continuous	Default	Default	Yes
SM6	**LOCK indicator of Forward Channel** (LOCK pin asserted in case of timing errors or no lock at the deserializer side)	PLL Lock error Timing errors	<10us	continuous	Default	Default	No
SM7	**ERRB indicator of Forward Channel** (In case of CRC errors, they are flagged on the deserializer ERRB output)	Forward data transmission errors	<10us	continuous	Default	Default	Yes
SM8	**LOCK indicator of Reverse Channel** LOCK pin asserted in case of timing errors, Continuous Mode. A “no LOCK” at the receiver side will also be	PLL Lock error Timing errors	<10us	continuous	Optional	Optional	No

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	flagged at the transmitter (serializer).						
SM9	**ERRB indicator of Reverse Channel** (ERRB pin asserted in case of reverse data transmission errors)	Reverse data transmission errors	<10us	continuous	Optional	Optional	Yes
SM10	**Error generator (serializer)** Fault injection generator at serializer side	Error in CRC generator	<10us	startup	Optional	Optional	Yes
SM11	**Video Sequence Number** There is an alternating one bit video packet sequence number to detect a video packet dropped and fill-in to sustain continuity. Error will be flagged through CRC. This sequence number is also valid for the control channel**.**	Dropped video packets Dropped control packets	<100us	continuous	Optional	Optional	No
SM12	**Acknowledge of UART or I2C**	Peer failure	<10us	continuous	Default	Default	Yes (indirectly through retransmissio n err)
SM14	**Eye Opening Monitor** Measure the link budget. This directly correlates with the bit error rate (see BER testing)	Loss of lock Cable signal integrity	<10us after EOM detection	continuous	Optional	Optional	Yes
SM15	**OV/UV comparators**	Over and Undervoltage on power supplies	<10us	continuous	Default	Default	Yes
SM16	**Evaluate CSI-2 ECC/CRC** CSI 2 CRC and ECC is evaluated at serializer input and deserializer output, and a register is set in case of failed CRC check	Data corruption in received CSI 2 image data. Checked at serializer input, and in case of video transmission by CSI-2 forwarding also checked at deserializer output.	< 50us	continuous	N/A	Optional	Yes
SM17	**FIFO Overflow Detection**	Overflow on backtop line memories	<300us (8k addr x pclk + buffer)	Continuous	Optional	Optional	No
SM18	**Logic BIST** Logic BIST is run as part of startup sequence.	Stuck-at faults in logic	< 10 ms	startup	Default	Default	No
SM19	**Memory BIST** Memory BIST is run as part of startup sequence	Stuck-at faults in memories	< 10 ms	Startup	Default	Default	No
SM20	**End-to-end CRC during video transmission using CSI-2 forwarding**	Data corruption in CSI-2 image data	< 50us (at DES checkpoint)	Continuous, On demand (evaluated at application level)	N/A	Default	N/A

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SM21	**CRC on I2C and UART transactions** Each I2C and UART transaction has a CRC associated with it. Upon detection of CRC error, command is not executed, and transaction is NACKed	Corrupted data being written to serializer I2C and/or UART interface	< 100us	Continuous	Optional	Optional	Yes
SM22	**Message counter** A message counter is incremented for each I2C transaction. A copy of the counter is maintained in both ends of the I2C link. If a mismatch is found, the transaction is rejected.	Missing or repeated I2C transaction	< 100us	Continuous	Optional	Optional	Yes
SM23	**GPO Readback:** GPO pin state can be read back and compared to last data written to the GPO	GPO driver output open or stuck	< 100us per GPIO	On demand	Default	Default	No
SM26	**Memory ECC**	Soft and hard memory faults	< 50us	Continuous	Default	Default	Yes
SM27	**Configuration register CRC**	Corrupted data in configuration registers	< 2.1 ms (if programmed to re-evaluate every 2ms)	Continuous	Default	Default	Yes
SM29	**CRC of non-volatile memory**	Stuck-at fault in programmable fuse array	< 2 ms	Startup	Default	Default	Yes
SM30	**Self-clearing unlock register**	Invalid attempt at re-configuration of serializer registers	< 100us	Continuous	Optional	Optional	No
SM32	**Running disparity check and illegal 9b/10b symbol check**	Decoding errors. Excessive running disparity, presence of invalid symbol.	<10us	Continuous	Optional	Optional	Yes (decode and idle errors)
SM34	**DPLL lock detector**	DPLL unlocked, outputting incorrect frequency	<100us	Continuous	Default	Default	No
SM35	**CSI-controller and D-PHY handshake**	If handshake is broken, backtop memories would overflow and error would be reported	<300us (8k addr x pclk + buffer)	Continuous	Optional	Optional	No
SM36	**Video lock**	Loss of video synchronization	<100us	Continuous	Default	Default	No
SM37	**GPIO open detection** Internal pull-ups and pull-downs are exercised in a sequential manner to detect if input is driven	GPIO input is not driven	<100us per GPIO	On demand	Default	Default	No
SM38	**Evaluate CSI-2 CRC** CSI 2 CRC and ECC is evaluated at serializer input and deserializer output, and a register is set in case of failed CRC check	CRC error	<50us	Continuous	Optional	Optional	Yes

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SM40	**Packet Counter for GPIO, I2C, SPI and UART packet** verify correct generation of control data packets and sequence	Packet Counter for I2C, UART, GPIO, SPI	<10us	On demand (host driven feature)	Optional	Optional	Yes
SM41	**Readback of status for ERRB and LFLT through I2C or UART register**	ERRB and LFLT	<100us each	On demand	Default	Default	N/A

Table 9: List of Safety Mechanisms for MAX96714

Note 1: It is the responsibility of the integrator to enable all the specified safety mechanisms to achieve ASIL B for the random hardware fault metrics of this specific IC.

Note 2: Safety Mechanisms SM38,SM39,SM40 and SM41 have been added during the FMEDA process to achieve compliance for the random HW fault metrics.

Note 3: Safety Mechanisms rated as "Default" are always required to reach the ASIL B compliance level for the IC and are associated as "optimized set". Using "default" mechanisms only will lead to optimized diagnostic times.

Note 4: Safety Mechanisms rated as "Optional" can be additionally enabled if desired. Using additional mechanisms will lead to increased diagnostic coverage. If all "Optional" safety mechanisms are enabled, "Baseline" FMEDA coverage can be used.

Note 5: With respect to ASIL metrics in pixel mode (SF1) and tunneling mode (SF2) refer to the FMEDA summary in section 10.2 and associated parameters for those individual system functions.

Note 6: The following Safety Mechanisms have been removed compared to the previous version of the Safety Manual: SM24 (overtemperature warning) and SM31 (exterior triggering of interrupt)

9. Verification of ASIL level

9.1. FMEDA of MAX96714

The FMEDA for MAX96714 has been performed considering all relevant functional blocks and is described in the FMEDA document.

For all used components within the element MAX96714 attributed for Function 1 *[Function2 etc.]*, the individual malfunctions are defined

- The Root Causes of those mal functions are identical to the FMEDA Failure Modes
- A failure split is defined with respect to the Root Causes for a malfunction considering different probabilities
- The corresponding diagnostic functions (safety measures) are defined to detect a specific Root Cause/FMEDA Failure Mode
- The Failure Rates (FIT Rates) are defined for the individual Components / Functional Blocks of the element MAX96714
- Furthermore, the diagnostic coverage is defined for any safety relevant Failure Mode
- Single point as well as latent failures has been considered.

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In this Safety Manual FMEDA examples for a functional block (*"GMSL2 Transmitter)" ar*e provided. The complete FMEDA includes all Functional Blocks according to the Technical Safety Concept.

FMEDA Example for Block FB64

To present the FMEDA for one Block, the following Table is used, which is part of the more detailed FMEDA table, which includes also Latent Fault section and Metrics per System Function as well as the pin level portion.

Single Point Fault Section														
Component Block Name	Component Function	Block Failure Rate / FIT	Relevant Block Failure Rate for FIT Rate calculation (internal)	Safety Related Hardware Component (SRC) or	Failure Modes	Effect of Failure Mode	SRC - Block FIT	NSR FIT in SRC Element (internal)	SRC FIT in SRC Elements (internal)	Failure rate Distribution (%)	Failure mode that has a potential to violate the safety goal in absence of Safety Mechanisms	Safety mechanism(s) allowing to prevent the failure mode from violating the safety goal	Failure mode coverage wrt. violation of safety goal	Assessed or Single Point Fault fau
FB64	MIPI DPhy Tx0 - DPLL	1505086539	1505086539	SRC	Output stuck high-low	No MIPI CSI 2 transmitted on Output lane	1505086539	0	0.376276	25	x	SM20, SM38	99	0.00376272
			1505086539		Incorrect gain/vOffset	Corrupted MIPI CSI 2 - transmitted output lane	0	0	0.376276	25	x	SM20, SM38	99	0.00376272
			1505086539	100	Wrong frequency - DPLL	Corrupted MIPI CSI 2 - transmitted output lane	0	0	0.7525433	50	x	SM34, SM20	99	0.00752543

Table 9: FMEDA – Example (Single Point Section)

9.2. Verification results of Safety Measures/HW Integration Testing

Based on the Hardware Design Specification, Technical Safety Concept and Validation Plan, the functionality of the Product MAX96714 is evaluated with special consideration on Safety Measures. The results of the verification of Safety Measures for MAX96714 are shown in the table below.

Safety Mechanism	Safety Mechanism Name	Failure Mode to detect	Test Case Description	Test Case Type	Test Method	Pass/Fail
SM1	**BIST for Serial Link (BER Testing using PRBS)**	Bit Error on Forward Video (Forward Video PRBS)	Bit Error on Forward Video (Forward Video PRBS) --- **3G NRZ**	Fault Injection Data	Functional Test and Software Test	Pass
		Bit Error on Forward Video (Forward Video PRBS)	Bit Error on Forward Video (Forward Video PRBS) --- **6G NRZ**	Fault Injection Data	Functional Test and Software Test	Pass
		Bit Error on Forward Video (Forward Video PRBS)	Bit Error on Forward Video (Forward Video PRBS) --- **GMSL1**	Fault Injection Data	Functional Test and Software Test	Pass
		Bit Error on GMSL link (GMSL Link PRBS)	Bit Error on GMSL link (GMSL Link PRBS) **--- 3G NRZ**	Fault Injection Data	Functional Test and Software Test	Pass
		Bit Error on GMSL link (GMSL Link PRBS)	Bit Error on GMSL link (GMSL Link PRBS) **--- 6G NRZ**	Fault Injection Data	Functional Test and Software Test	Pass
		Bit Error on Forward Video (Forward Video PRBS)	Bit Error on GMSL link (GMSL Link PRBS) --- **GMSL 1**	Fault Injection Data	Functional Test and Software Test	Pass
SM2	**Line Fault Detection on Serializer Output**	SIOA+ Short to Battery	The local side checks for short to battery on SIOA+, indication is through pin and registers.	Fault Injection Hardware	Functional Test	Pass

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		SIOA- Short to Battery	The local side checks for short to battery on SIOA- , indication is through pin and registers.	Fault Injection Hardware	Functional Test	Pass
		SIOA+ Short to GND	The local side checks for short to GND is on SIOA+, indication is through pin or register.	Fault Injection Hardware	Functional Test	Pass
		SIOA- Short to GND	The local side checks for short to GND is on SIOA-, indication is through pin or register.	Fault Injection Hardware	Functional Test	Pass
		SIOA+ Line Open	The local side checks for open line is on SIOA+, indication is through pin or register.	Fault Injection Hardware	Functional Test	Pass
		SIOA- Line Open	The local side checks for open line is on SIOA-, indication is through pin or register.	Fault Injection Hardware	Functional Test	Pass
		SIOA+/SIOA- Line-to-Line Short	The local side checks for Line-to-Line short between SIOA+ and SIOA-, indication is through pin or register.	Fault Injection Hardware	Functional Test	Pass
SM3	**CRC for Control Channel**	Forward GPIO CRC	Forward GPIO CRC	Fault Injection Software	Software Test	Pass
		Reverse GPIO CRC	Reverse GPIO CRC	Fault Injection Software	Software Test	Pass
		Forward control I2C CRC	Forward control I2C CRC	Fault Injection Software	Software Test	Pass
		Reverse Control I2C CRC	Reverse Control I2C CRC	Fault Injection Software	Software Test	Pass
		Forward Control UART CRC	Forward Control UART CRC	Fault Injection Software	Software Test	Pass
		Reverse Control UART CRC	Reverse Control UART CRC	Fault Injection Software	Software Test	Pass
SM4	**CRC for Forward Video Channel**	Bit error on Forward Video (Line CRC) -- **pixel mode**	Line CRC toggling to verify Forward Video	Fault Injection Software	Software Test	Pass
		Bit error on GMSL Link (Link CRC) -- **pixel mode**	GMSL Link CRC enabling and disabling to verify GMSL Link	Fault Injection Software	Software Test	Pass
		Bit error on GMSL Link (Link CRC) -- **tunneling mode**	GMSL Link CRC enabling and disabling to verify GMSL Link	Fault Injection Software	Software Test	Pass
SM5	**Error Correction through Re-Transmission (Correction of up to 16 bit errors in each packet)**	Error in control channel	Check that re-transmission occurs	Fault Injection Software	Software Test	Pass
SM6	**Lock Indicator of Forward Channel**	Loss of GMSL Forward Link	Verify Lock, reconfigure Serializer to verify loss of lock.	Fault Injection Software	Software Test	Pass
SM7	**ERRB Indicator of Forward Channel**	Local Error in GMSL Link	Verify local error	Fault Injection Software	Software Test	Pass
SM8	**Lock Indicator of Reverse Channel**	Loss of GMSL Reverse Link	Verify Lock, reconfigure Serializer to verify loss of lock.	Fault Injection Software	Software Test	Pass

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SM9	**ERRB Indicator of Reverse Channel**	Reverse channel transmission errors	Verify remote error	Fault Injection Software	Software Test	Pass
SM10	**Error Generator**	Decoding Errors	Check decoding errors	Fault Injection Software	Software Test	Pass
SM11	**Video Sequence Numbering**	Video Packet Loss	Dropped video packets Dropped control packets Test in pixel mode and tunneling mode	Fault Injection Software	Software Test	Pass
SM12	**Acknowledge of UART or I2C**	Peer Failure	Standard feature of UART	ACK is received	Software and Hardware Test	Pass
		Peer Failure	Standard feature of I2C	ACK is received	Software and Hardware Test	Pass
SM14	**Eye Opening Monitor ----- 3G and 6G NRZ**	Visual verification of **3GHz NRZ** eye	Use GMSL GUI eye mapper	Fault Injection Software	Functional Test	Pass
		Visual verification of **6GHz NRZ** eye	Use GMSL GUI eye mapper	Fault Injection Software	Functional Test	Pass
SM15	**OV/UV comparators**	Over/Undervoltage of VDD, Undervoltage of IOVDD, VDD18	Manually adjust voltage levels, read I2C to read message/detect lockout	Software and Hardware Test	Functional Test	Pass
SM16	**Evaluate CSI-2 ECC/CRC in Tunneling mode**	CRC errors	Check CRC errors	Fault Injection Software	Functional Test	Pass
SM17	**FIFO Overflow Detection**	Video Overflow	Same as SM35	N/A	N/A	Pass
SM18	**Logic BIST**	Stuck-at errors in logic on startup	Used to check internal logic gates are able to toggle.	Fault Injection Software	ATE test	Pass
SM19	**Memory BIST**	Bit errors in memories on startup	Used to check internal memory bits are able to toggle. Can be run either POST or LBIST. LBIST can be run through TM's.	Fault Injection Software	ATE Test	Pass
SM20	**End-to-end CRC**	Bit errors in video transmission Note: Error is detected at host. Requires host to check CRC upon reception of CSI-2 packet.	Error injection of CSI video source: Verify SER records CRC error and DES should just forward corrupted CSI data. Analyzer should detect CRC error. **-- DPHY**	Fault Injection Software	Functional and Software test	Pass
			Send good CSI data. SER and DES should preserve ECC/CRC. Verify CRC is correct using MIPI analyzer. **-- DPHY**	Fault Injection Software	Functional and Software test	Pass
SM21	**CRC on I2C and UART transactions**	Bit errors in I2C and UART transactions (host interface)	Build I2C driver to generate normal I2C with addition of CRC and message counter. Using description in IOS, create proper CRC. Verify serializer/deserializer are able to detect the correct CRC.	Fault Injection Software	Functional and Software test	Pass
			Build I2C driver to generate normal I2C with addition of CRC and message counter. Using description in IOS, create improper CRC. Verify	Fault Injection Software	Functional and Software test	Pass

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			serializer/deserializer the improper CRC.			
			Build UART driver to generate normal UART with addition of CRC and message counter. Using description in IOS, create proper CRC. Verify serializer/deserializer are able to detect the correct CRC.	Fault Injection Software	Functional and Software test	Pass
			Build UART driver to generate normal UART with addition of CRC and message counter. Using description in IOS, create improper CRC. Verify serializer/deserializer the improper CRC.	Fault Injection Software	Functional and Software test	Pass
			Build I2C driver to generate normal I2C with addition of CRC and message counter. Using description in IOS, create improper CRC. Verify serializer/deserializer detect proper number of CRC errors sent.	Fault Injection Software	Functional and Software test	Pass
SM22	**Message counter**	Missing or repeated I2C transaction (host interface)	Build I2C driver to generate normal I2C with addition of CRC and message counter. Using description in IOS, create improper message count. Verify serializer/deserializer the proper message count.	Fault Injection Software	Functional and Software test	Pass
			Build I2C driver to generate normal I2C with addition of CRC and message counter. Using description in IOS, create improper message count. Verify serializer/deserializer the improper message count.	Fault Injection Software	Functional and Software test	Pass
			Build UART driver to generate normal UART with addition of CRC and message counter. Using description in IOS, create improper message count. Verify serializer/deserializer the proper message count.	Fault Injection Software	Functional and Software test	Pass
			Build UART driver to generate normal UART with addition of CRC and message counter. Using description in IOS, create improper message count. Verify serializer/deserializer the improper message count.	Fault Injection Software	Functional and Software test	Pass

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			Build I2C driver to generate normal I2C with addition of CRC and message counter. Using description in IOS, create improper message count. Verify serializer/deserializer the proper message count. Verify able to reset message counter in SERDES.	Fault Injection Software	Functional and Software test	Pass
			Build I2C driver to generate normal I2C with addition of CRC and message counter. Using description in IOS, create improper message count. Verify serializer/deserializer the improper message count. Verify SERDES counts correctly the number of err'd messages.	Fault Injection Software	Functional and Software test	Pass
SM23	**Readback of logic state for GPIO**	Readback Error	Hardware test GPIO state	Fault Injection Software	Functional Test	Pass
SM26	**Memory ECC**	Bit error in volatile memory	Use MEM ECC error injection to inject errors in the memory	Fault Injection Software	Functional and Software test	Pass
SM27	**Configuration register CRC**	Configuration register stuck or inadvertently changed for Des	1) Configure parts 2) Enable "Config reg CRC" for Des 3) Change any configuration Register bit value 4) ERRB asserted, dedicated register set on Des	Fault Injection Software	Software and Hardware Test	Pass
SM29	**CRC of non-volatile memory**	Bit error in non-volatile memory	Design Simulation	Fault Injection Software	Software Simulation	Pass
SM30	**Self-clearing unlock register**	Invalid attempt at re-configuration of serializer registers	Error injection	Fault Injection Software	Functional and Software test	Pass
SM32	**Running disparity check and illegal 9b/10b symbol check**	9b/10b encoding errors are detected through running disparity check	Link Failure -- GMSL2 mode	Fault Injection Software	Functional and Software test	Pass
			Link Failure – GMSL1 mode	Fault Injection Hardware	Functional and Software test	Pass
SM33	**Clock Monitor: 150 MHz clock compared against crystal clock, Leading to LOCK error**	Clock Failure	CMU_LOCK failure by increasing/decreasing the external clock	Fault Injection Hardware	Functional and Software test	Pass
SM34	**DPLL lock detector**	MIPI DPHY PLL, Wrong Frequency	Check the DPLL Lock register for the CSI ouput.	Fault Injection Hardware and Software	Software Test	Pass
SM35	**CSI-controller and D-PHY handshake**	Wrong Conversion of Pixel Data to CSI 2 /Stuck High Low -- **pixel mode**	1) Send MIPI data from Ser to Des 2) Set slower speed on Des (reg 0x320) 3) Check the overflow reg 0x312 for line overflow	Fault Injection Software	Functional and Software test	Pass

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			and command overflow errors			
		Wrong Conversion of Pixel Data to CSI 2 /Stuck High Low -- **tunneling mode**	1) Send MIPI data from Ser to Des 2) Set slower speed on Des (reg 0x320) 3) Check the overflow reg 0x312 for line overflow and command overflow errors	Fault Injection Software	Functional and Software test	Pass
SM36	**Video lock**	Reception of corrupted video data -- **pixel DPHY mode**	Check the VIDEO_LOCK register	Fault Injection Hardware and Software	Software Test	Pass
		Reception of corrupted video data -- **Tunneling DPHY mode**	Check the VIDEO_LOCK register	Fault Injection Hardware and Software	Software Test	Pass
SM37	**GPIO open detection**	GPIO open	1) Turn on 1Mohm pull up 2) Read back high 3) Turn on 1Mohm pull down instead now 4) Read back low 5) If both of these happen, then this means this pin is not driven externally because even with very weak drive we are able to change the state of the pin to both high and low	Fault Injection Hardware and Software	Functional and Software test	Pass
SM38	**Evaluate CSI-2 CRC**	CRC error	Identical to SM16, no additional testing needed.	NA	NA	Pass
SM40	**Packet Counter for GPIO, I2C, SPI and UART packet**	Packet Counter for I2C, UART, GPIO	Check PKT_CNT register and make sure it is changing	Fault Injection Hardware and Software	Functional and Software test	Pass
SM41	**Readback of status for ERRB and LFLT through I2C or UART register**	ERRB and LFLT	Read ERR status and LFLT registers using I2C/UART. Verified in SM2, SM7, SM9	NA	NA	Pass

Table 8: Verification results of Safety Measures for MAX96714

9.3. Summary of Safety Analyses/FMEDA

In order to verify the hardware related metrics according to ISO26262-Part 5, dedicated Safety Analyses/FMEDA has been performed. As a result of the FMEDA, an ASIL summary is created and contains the following information:

- Summary of the random HW fault Metrics for the IC (SFPM, LFM, PMHF)
- Summary of the random HW-Metrics for every System Function.

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FMEDA Summary – Baseline:

Note: for the baseline it is assumed that all Safety Mechanisms according to Table 6 are enabled (both default and optional mechanisms). This set will provide highest diagnostic coverage and low failure rate.

SPFM	LPFM	ASIL	FIT Rate - Whole IC	Residual FIT Single Point	Residual FIT Latent	PMHF
96.22%	92.41%	B	53.82446605	2.036550324	3.931977579	2.038524915

Table 9: FMEDA Summary for the whole IC (baseline)

System Function	System Function Description	SPFM	Residual FIT Single Point	ASIL Level
SF1	Video Transmission in Pixel Mode	96.64%	1.717290648	B
SF2	Video Transmission by CSI-2 Forwarding	97.42%	1.296063673	B
SF3	Route Control Data Over Forward channel	96.96%	1.329477947	B
SF4	Route Control Data Over Reverse Channel	96.81%	1.322179594	B

Table 10: FMEDA Summary per System Function and Safety Goal (baseline)

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FMEDA Summary – Optimized Set:

Note: for the optimized set it is assumed that only Safety Mechanisms rated as “Default” according to Table 6 are enabled. This set will provide optimized diagnostic time.

SPFM	LPFM	ASIL	FIT Rate - Whole IC	Residual FIT Single Point	Residual FIT Latent	PMHF
91.55%	91.44%	B	53.82446605	4.549539329	4.216522302	4.551687287

Table 11: FMEDA Summary for the whole IC (optimized list of SMs)

System Function	System Function Description	SPFM	Residual FIT Single Point	ASIL Level
SF1	Video Transmission in Pixel Mode	95.07%	2.515495236	B
SF2	Video Transmission by CSI-2 Forwarding	95.83%	2.094268261	B
SF3	Route Control Data Over Forward channel	91.88%	3.545028047	B
SF4	Route Control Data Over Reverse Channel	90.76%	3.8351686	B

Table 12: FMEDA Summary per System Function and Safety Goal (optimized list of SMs)

The FMEDA results show compliance of the random HW faults metrics according to ASIL B requirements.

Note on FMEDA: the base failure rate has been determined according to SN29500 standard at a virtual junction temperature of T_j=55 deg.C

Note on Safety Analysis: Further Safety Analyses (like Dependent Failure Analysis) have been performed and may be reviewed on request.

Appendix

Rating Tables for Severity, Controllability and Exposure

Class	S0	S1	S2	S3
Description	No injuries	Light and moderate Injuries	Severe and life-threatening injuries (survival probable)	Life-threatening injuries (survival uncertain), fatal injuries

Table 2: Classes of Severity

Class	C0	C1	C2	C3
Description	Controllable in general	Simply controllable	Normally controllable	Difficult to control or uncontrollable

Table 3: Classes of Controllability

Class	E0	E1	E2	E3	E4
Description	Incredible	Very low probability	Low probability	Medium probability	High probability

Table 4: Classes of probability of exposure regarding operational situations

ASIL Classification Table

Severity class	Probability class	C1	C2	C3
S1	E1	QM	QM	QM
	E2	QM	QM	QM
	E3	QM	QM	A
	E4	QM	A	B
S2	E1	QM	QM	QM
	E2	QM	QM	A
	E3	QM	A	B
	E4	A	B	C
	E1	QM	QM	A

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S3	**E2**	QM	A	B
	E3	A	B	C
	E4	B	C	D

Table 5: ASIL determination

ASIL A - lowest safety integrity level
ASIL D - highest safety integrity level.
QM (quality management) - no requirement to comply with ISO 26262

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Terms and Definitions

ASIL (Automotive Safety Integrity Level)
One of four levels to specify the item's or element's necessary requirements of ISO 26262 and safety measures to apply for avoiding an unreasonable residual risk with D representing the most stringent and A the least stringent level.

Component
Non-system level element that is logically and technically separable and is comprised of more than one hardware part or of one or more software units.

Controllability (C)
Ability to avoid a specified harm or damage through the timely reactions of the persons involved, possibly with support from external measures.
Note: Persons involved can include the driver, passengers or persons in the vicinity of the vehicle's exterior.

Diagnostic coverage
Proportion of the hardware element failure rate that is detected or controlled by the implemented safety mechanisms.
Note: Diagnostic coverage can be assessed with regard to residual faults or with regard to latent multiple-point faults that might occur in a hardware element.

Element
System or part of a system including components, hardware, software, hardware parts, and software units.

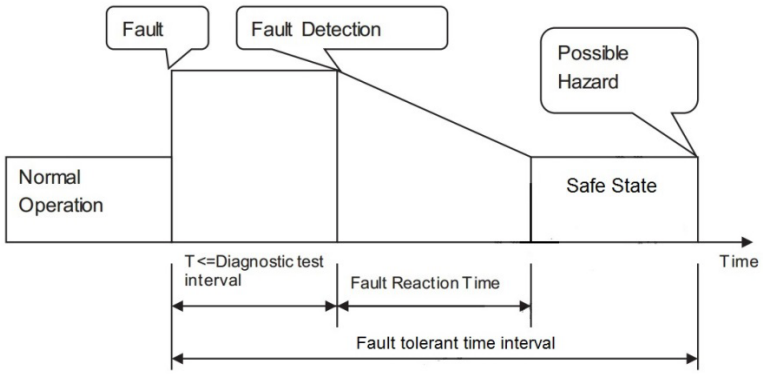
Exposure (E)
State of being in an operational situation that can be hazardous if coincident with the failure mode under analysis.

Failure rate
Probability density of failure divided by probability of survival for a hardware element.
Note: The failure rate is assumed to be constant.

FMEDA Failure Modes Effects and Diagnostic Analysis
Procedure for the detailed determination of error causes and their impact on the system and is used in the early stages of systems development for the purpose of early identification of weaknesses. The requirements of safety standards are the avoidance of errors within a safety-relevant system and to reduce them to a reliable residual error probability or a residual error rate.

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Fault reaction time and fault tolerant time interval



Harm
Physical injury or damage to the health of persons.

Hazard
Potential source of harm caused by malfunctioning behavior of the item.

Hazard analysis and risk assessment
Method to identify and categorize hazardous events of items and to specify safety goals and ASILs related to the prevention or mitigation of the associated hazards in order to avoid unreasonable risk.

Hazardous event
Combination of a hazard and an operational situation.

Item
System or array of systems to implement a function at the vehicle level, to which ISO 26262 is applied.

Latent fault
Multiple-point fault whose presence is not detected by a safety mechanism nor perceived by the driver within the multiple-point fault detection interval.

Operational situation
Scenario that can occur during a vehicle's life. For example: driving, parking, maintenance.

Residual fault
Portion of a fault that by itself leads to the violation of a safety goal, occurring in a hardware element, where that portion of the fault is not covered by safety mechanisms.
Note: This presumes that the hardware element has safety mechanism coverage for only a portion of its faults.
Example: If low (60 %) coverage is claimed for a failure mode, the other 40 % of that same failure mode is the residual fault.

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Residual Risk
Risk remaining after the deployment of safety measures.

Risk
Combination of the probability of occurrence of harm(1.56) and the severity(1.120) of that harm.

Safety goal
Top-level safety requirement as a result of the hazard analysis and risk assessment.
Note: One safety goal can be related to several hazards, and several safety goals can be related to a single hazard.

Safety measures
Activity or technical solution to avoid or control systematic failure and to detect random hardware failures or control random hardware failures, or mitigate their harmful effects
Note 1: Examples of safety measures are FMEA (Failure Mode and Effects Analysis) and software without the use of global variables.
Note 2: Safety measures include safety mechanisms.

Safety mechanism
Technical solution implemented by E/E functions or elements, or by other technologies, to detect faults or control failures in order to achieve or maintain a safe state.
Note 1 Safety mechanisms are implemented within the item to prevent faults from leading to single-point failures or to reduce residual failures and to prevent faults from being latent.
Note 2 The safety mechanism is either
a) able to transition to, or maintain, the item in a safe state, or
b) able to alert the driver such that the driver is expected to control the effect of the failure,
as defined in the functional safety concept.

Severity (S)
Estimate of the extent of harm to one or more individuals that can occur in a potentially hazardous situation.

Single-point fault
Fault in an element that is not covered by a safety mechanism and that leads directly to the violation of a safety goal.

System
Set of elements that relate at least a sensor, a controller and an actuator with one another. *[Summary for all ISO-related terms and definitions to be added here.]*

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REVISION HISTORY

REV	CHANGES MADE	DATE	INIT.
A	CN. 1116571 Initial release. Submission of Safety Manual	03/05/21	TJH
B	CN 1127115 CSA add Optimized SM list	05/26/21	CSA
C	CN 1148124. Remove SM24 and SM31 (not in the FMEDA)	1/6/22	CS
D	CN 1155699. D: Revise statement in section 8 (remove option to stop video transmission) Make SM37 required in both pixel and tunnel mode (fix typo)	3/22/22	CS

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